

MIDWAY ISLAND, United States Minor Outlying Islands

WMO: 910660

Lat: 28.202N

Lon: 177.383W

Elev: 18

StdP: 14.69

Time Zone: -11.00 (W11)

Period: 98-19

WBAN: 22701

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	57.8	59.5	41.8	39.0	60.7	44.3	42.9	62.1	32.9	66.3	29.3	66.3	9.1	340	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	87.9	77.9	86.3	77.1	85.2	76.5	79.3	84.5	78.4	83.7	77.9	83.2	10.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.9	144.9	82.2	76.9	140.2	81.9	76.0	136.1	81.5	42.5	84.5	41.6	84.1	40.8	83.2	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.7	23.8	20.6	DB	55.2	89.5	2.2	4.2	53.6	92.6	52.3	95.0	51.1	97.4	49.5	100.5
			WB	48.2	81.0	1.4	1.8	47.2	82.2	46.4	83.3	45.6	84.3	44.6	85.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.8	67.5	66.8	67.8	69.8	72.4	77.2	79.9	81.4	80.7	77.5	73.5	70.3
	DBStd	6.03	3.69	3.93	3.65	3.28	3.69	2.77	1.80	2.03	2.25	2.92	3.06	3.25
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	61	18	24	14	3	0	0	0	0	0	0	0	2
	CDD50	8675	543	470	550	594	695	815	928	974	921	851	705	629
	CDD65	3264	96	75	100	146	230	365	463	509	471	386	256	167
	CDH74	21537	68	75	109	202	740	2478	4233	5320	4663	2670	766	213
	CDH80	4053	5	4	7	3	50	302	820	1409	1097	329	24	3
(14) Wind	WSAvg	10.8	13.1	12.0	11.5	10.7	9.9	9.6	10.8	10.0	8.9	10.0	11.5	11.4
(15) Precipitation	PrecAvg	41.3	5.3	3.5	3.8	2.5	1.6	2.0	3.8	4.6	4.3	3.6	4.5	4.3
	PrecMax	58.1	15.6	8.1	10.1	7.1	7.2	7.7	13.3	12.1	15.9	10.0	12.0	17.8
	PrecMin	21.0	1.7	0.9	0.0	0.1	0.2	0.3	0.6	0.4	1.1	0.3	0.8	0.9
	PrecStd	10.2	2.9	2.1	2.9	2.0	1.5	1.6	2.8	3.3	3.2	2.9	3.1	3.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.1	80.7	80.9	80.7	83.4	86.4	87.9	89.7	89.8	86.2	82.2	80.7
		MCWB	69.1	69.6	70.0	71.3	74.9	77.0	77.3	78.5	78.7	77.6	73.5	70.9
	2%	DB	75.4	76.0	76.7	78.2	81.4	84.1	86.0	88.0	87.8	84.3	80.9	78.5
		MCWB	69.3	68.0	68.8	70.4	72.9	75.7	76.3	78.0	77.9	76.1	73.0	70.8
	5%	DB	74.1	73.8	74.6	76.2	79.8	82.6	84.6	86.5	86.0	83.1	79.4	76.2
		MCWB	68.5	67.3	68.1	69.4	71.3	74.9	75.8	77.3	77.1	75.3	72.1	70.0
	10%	DB	73.0	72.6	72.9	74.8	78.5	81.8	83.8	85.4	84.5	81.9	78.0	74.9
		MCWB	67.7	66.8	67.3	68.5	71.0	74.5	75.5	76.8	76.5	74.6	71.4	69.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	72.6	72.8	74.2	76.6	78.5	78.9	80.3	80.3	80.7	77.1	74.4
		MCDB	75.2	75.8	75.9	76.1	80.3	83.1	83.6	85.8	85.7	82.4	80.5	77.3
	2%	WB	71.8	70.8	71.3	72.8	75.1	77.5	77.9	79.1	79.2	78.7	75.4	73.1
		MCDB	74.1	74.1	74.2	75.5	79.2	81.5	83.1	85.1	85.0	81.8	78.7	75.6
	5%	WB	70.6	69.4	70.2	71.7	73.9	76.7	77.2	78.3	78.3	77.2	74.3	71.9
		MCDB	73.1	72.6	73.1	74.7	77.7	80.8	82.7	84.3	84.0	81.2	77.8	74.7
	10%	WB	69.3	68.1	68.9	70.5	72.9	75.9	76.5	77.7	77.6	76.0	73.0	70.9
		MCDB	72.1	71.5	72.0	73.7	76.7	80.0	82.0	83.7	83.2	80.7	76.7	74.0
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	7.8	7.5	7.6	7.5	7.4	7.8	7.9	7.8	7.6	7.1	7.2
		MCDBR	8.8	9.2	8.7	8.3	8.4	8.1	8.5	8.8	9.1	8.2	8.0	8.2
	5% WB	MCWBR	8.0	7.5	6.7	5.7	5.2	4.3	4.6	4.6	4.8	4.8	5.8	6.3
		MCWBR	8.3	9.1	8.2	8.0	8.0	7.5	8.1	8.0	8.1	7.4	7.4	7.6
(40) Clear-Sky Solar Irradiance	taub		0.356	0.364	0.385	0.412	0.410	0.393	0.379	0.378	0.376	0.369	0.346	0.356
		taud	2.571	2.542	2.470	2.402	2.456	2.546	2.608	2.620	2.613	2.616	2.665	2.604
	Ebn at Noon		282	288	287	281	280	283	287	287	286	283	284	277
		Edh at Noon	28	30	35	38	36	33	31	30	30	28	25	26
(44) All-Sky Solar Radiation	RadAvg		1035	1298	1546	1900	2061	2208	2256	2064	1840	1507	1133	988
	RadStd		83	90	102	106	130	134	118	125	91	98	73	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page